The Jeans Criterion in Cosmological AMR Simulations: Cooling, Holdback, and Refinement

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ABSTRACT

The Jeans length criterion is a fundamental numerical stability requirement in self-gravitating hydrodynamic simulations: failing to resolve it by at least four cell widths produces artificial fragmentation that contaminates the star formation history. In cosmological adaptive mesh refinement (AMR) simulations, however, three distinct mechanisms conspire to violate or misapply this criterion. First, the *cosmological holdback* that prevents premature refinement at high redshift blocks the Jeans criterion along with all other refinement triggers, creating abrupt starbursts when levels are released. Second, the standard Newton–Raphson cooling solver can overshoot the thermal equilibrium temperature $T_{\rm eq}$, introducing transient errors in the Jeans length evaluation. Third, pre-computed cooling tables at a single redshift become stale as the UV background evolves. We address all three issues in the cuRAMSES code: (i) a *Jeans-independent holdback bypass* that allows the Jeans criterion to operate ahead of the holdback, with a configurable level limit to prevent runaway cascades; (ii) an *exact cooling integrator* based on the Townsend (2009) Y-function method applied to three-dimensional Grackle equilibrium cooling tables, eliminating Newton–Raphson overshooting; and (iii) *multi-redshift table interpolation* that tracks the evolving UV background across cosmic time. We quantify how cooling rate differences between the standard RAMSES solver and the Grackle library propagate into 40–80 per cent errors in the Jeans length at low densities, and demonstrate that the exact integrator removes the stochastic refinement noise caused by temperature overshooting. Future extensions incorporating *turbulent pressure support* into the Jeans criterion are outlined.

Key words: methods: numerical – cosmology: simulations – stars: formation – hydrodynamics – instabilities – turbulence

1 INTRODUCTION

Gravitational instability is the fundamental driver of cosmic structure formation. Jeans (1902) showed that a uniform, self-gravitating medium is unstable to perturbations on scales exceeding the Jeans length

$$\lambda_{\rm J} = \sqrt{\frac{\pi\, c_s^2}{G\,\rho}}, \tag{1}$$

where c_s is the sound speed and ρ the gas density. All scales below $\lambda_{\rm J}$ are stabilised by thermal pressure; all scales above it collapse. In astrophysical simulations of star-forming gas, the Jeans length therefore defines a *minimum resolution requirement*: if the computational grid or particle spacing cannot resolve $\lambda_{\rm J}$, the simulation produces qualitatively wrong results — artificial fragmentation that bears no relation to the true physical outcome (Truelove et al. 1997; Bate & Burkert 1997).

Cosmological simulations of galaxy formation must follow gravitational collapse across many orders of magnitude in spatial scale, from ∼ 100 Mpc cosmic web filaments down to ∼ 10 pc star-forming clouds. Adaptive mesh refinement (AMR; Berger & Oliger 1984; Berger & Colella 1989) achieves this by hierarchically subdividing cells in regions that require higher resolution, concentrating computational resources where they are needed. Major AMR codes used in cosmological contexts include RAMSES (Teyssier 2002), Enzo (Bryan et al. 2014), and ART (Kravtsov, Klypin & Khokhlov 1997). Alternative numerical techniques — smoothed particle hydrodynamics (SPH; Springel 2005), moving-mesh methods (Springel 2010), and mesh-free finite-mass schemes (Hopkins 2015) — face analogous resolution requirements (Bate & Burkert 1997; Nelson et al. 2009), but the AMR framework provides a particularly transparent mechanism for enforcing the Jeans condition: a cell is flagged for refinement whenever $\lambda_{\rm J} < N_J\,\Delta x$, where N_J is typically 4–8 and Δx is the cell width (Truelove et al. 1997).

In practice, however, enforcing the Jeans criterion in cosmological AMR simulations is complicated by three interacting effects:

(i) Cosmological holdback. To prevent premature over-refinement at high redshift — which would exhaust memory before the epoch of interest — codes such as RAMSES employ a *holdback* mechanism that progressively releases refinement levels as the scale factor a grows. This holdback blocks *all* refinement criteria simultaneously, including the Jeans criterion, which is a numerical *stability* requirement rather than merely a resolution preference. When a level is finally released, a burst of refinement and star formation follows (see also Dubois et al. 2014, 2021).

(ii) Cooling accuracy. The Jeans length is proportional to $T^{1/2}$, so the equilibrium temperature $T_{\rm eq}$ of the gas directly determines whether refinement is triggered. $T_{\rm eq}$ is set by the balance between radiative cooling and photo-heating by the ultraviolet (UV) background (Haardt & Madau 2012; Faucher-Giguère 2020). Errors in the cool-

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ing rate tables (Sutherland & Dopita 1993; Ferland et al. 2017), the UV background model (Courty & Alimi 2004), or the numerical integration scheme all propagate into errors in $T_{\rm eq}$ and hence in $\lambda_{\rm J}$. The standard RAMSES cooling solver uses a semi-implicit Newton–Raphson (N-R) iteration that is known to overshoot $T_{\rm eq}$ when the net cooling rate changes sign (cf. Townsend 2009), introducing stochastic noise into the Jeans length evaluation. The Grackle equilibrium chemistry library (Smith et al. 2017) provides more accurate cooling rates — including density-dependent metal cooling and non-local thermodynamic equilibrium (non-LTE) molecular weights — but the standard tabulated approach still requires a robust numerical integrator to avoid overshooting.

(iii) Redshift evolution. The metagalactic UV background evolves strongly with redshift: it rises during cosmic reionisation ($z \gtrsim 6$), peaks near $z \sim 2$–3, and declines towards $z = 0$ (Haardt & Madau 2012; Faucher-Giguère 2020). A cooling table computed at a single redshift becomes increasingly inaccurate as the simulation evolves away from that epoch, producing systematic biases in $T_{\rm eq}$ and $\lambda_{\rm J}$ that can reach factors of 2–3 during the He II reionisation epoch ($z \sim 3$–4).

Beyond thermal pressure, *turbulence* provides an additional source of pressure support that can significantly modify the effective Jeans length. Chandrasekhar (1951) and Bonazzola et al. (1987) showed that turbulent motions increase the effective sound speed from c_s to $(c_s^2 + \sigma_{\rm turb}^2)^{1/2}$, where $\sigma_{\rm turb}$ is the turbulent velocity dispersion, yielding a turbulent Jeans length

$$\lambda_{\rm J}^{\rm turb} = \sqrt{\frac{\pi\,(c_s^2 + \sigma_{\rm turb}^2)}{G\,\rho}}. \tag{2}$$

In the interstellar medium (ISM), turbulence is driven by supernova feedback, gravitational infall, magnetorotational instability, and galactic shear (Robertson & Kravtsov 2008; Kim & Ostriker 2017; Hopkins et al. 2018). Subgrid-scale (SGS) turbulence models (Schmidt, Niemeyer & Hillebrandt 2006; Schmidt & Federrath 2011; Rennehan 2021) can estimate the unresolved turbulent energy on the grid scale, providing an in-situ value of $\sigma_{\rm turb}$ that can be fed into the refinement criterion. In the context of star formation, Federrath & Klessen (2012) showed that turbulence simultaneously *supports* gas on large scales (raising the effective Jeans mass) and *compresses* gas locally (triggering collapse in overdense regions), so the net effect on refinement is scale-dependent. Semenov, Kravtsov & Gnedin (2016) demonstrated that a turbulence-dependent star formation model produces more realistic star formation histories than density-threshold prescriptions. Incorporating turbulent pressure support into the AMR Jeans criterion — using either resolved velocity gradients or SGS models — is a natural extension that we plan to address in future work.

Large-volume cosmological simulations such as Horizon-AGN (Dubois et al. 2014), EAGLE (Schaye et al. 2015), IllustrisTNG (Pillepich et al. 2018), NewHorizon (Dubois et al. 2021), and Horizon Run 5 (HR5; Lee et al. 2021) rely on accurate Jeans length resolution to produce converged star formation rates and galaxy properties. Any systematic error in the Jeans criterion — whether from the holdback, cooling inaccuracies, or the neglect of turbulent pressure — propagates into the predicted stellar masses, morphologies, and feedback histories.

In this paper we present three improvements to the handling of the Jeans criterion in cosmological AMR simulations, implemented in CURAMSES:

(i) An *improved cooling module* that combines 3D Grackle equilibrium cooling tables with the Townsend (2009) exact integration scheme and multi-redshift interpolation, eliminating the N-R overshooting that contaminates the Jeans length near thermal equilibrium (Section 3).

(ii) A *Jeans-independent holdback bypass* that allows the Jeans criterion to operate ahead of the cosmological holdback, with a configurable level limit to prevent uncontrolled refinement cascades (Section 5).

(iii) Analysis of the *polytropic temperature floor* and its role in self-regulating the bypass cascade through physics-based density thresholds (Section 6).

We also outline future directions for incorporating turbulent pressure support into the Jeans criterion (Section 9.4).

The paper is organised as follows. Section 2 reviews AMR refinement in RAMSES, including the holdback mechanism. Section 3 analyses the cooling module and its impact on the Jeans length. Section 4 identifies the starburst problem caused by the holdback. Section 5 presents the Jeans bypass solution. Section 6 discusses the polytropic floor. Section 7 gives implementation details. Section 8 shows test results. Section 9 discusses implications, and Section 10 summarises our findings.

2 AMR REFINEMENT IN RAMSES

RAMSES (Teyssier 2002) is a grid-based AMR code that solves the coupled Euler–Poisson–Boltzmann equations on a fully threaded octree. Each cell at level $\ell < N_{\rm levelmax}$ is tested for refinement at every coarse time-step. The decision to subdivide a cell into $2^{N_{\rm dim}}$ children is governed by a combination of physical, geometric, and cosmological criteria.

2.1 Refinement Logic

The overall refinement flag for a cell is determined by the following Boolean expression:

$$\mathcal{R} = \neg\mathcal{H} \;\wedge\; \mathcal{G} \;\wedge\; \big(\mathcal{M} \vee \mathcal{S} \vee \mathcal{D} \vee \mathcal{P} \vee \mathcal{U} \vee \mathcal{J}\big), \tag{3}$$

where each symbol is defined below.

2.1.1 Physical Criteria (OR)

(i) **Quasi-Lagrangian mass criterion ($\mathcal{M}$).** The cell is flagged if its gas density exceeds a level-dependent threshold:

$$\rho_{\rm cell} \;\geq\; m_{\rm refine}(\ell)\,\frac{m_{\rm sph}}{V_{\rm cell}}, \tag{4}$$

where $m_{\rm sph}$ is the effective SPH kernel mass and $V_{\rm cell} = (\Delta x_\ell)^3$. This criterion ensures that regions containing roughly $m_{\rm refine}$ or more particles per cell are refined, providing a quasi-Lagrangian resolution that tracks gravitational collapse. Controlled by the namelist array `m_refine(1:nlevelmax)`.

(ii) **Sink particle forcing ($\mathcal{S}$).** If `sink_refine=.true.`, any cell containing a sink cloud particle (identified by `idp` < 0 and `tp` $= 0$) is unconditionally flagged up to $N_{\rm levelmax}$, guaranteeing maximum resolution around each accretion region.

(iii) **Density gradient ($\mathcal{D}$).** The normalised density gradient between a cell and each of its $2N_{\rm dim}$ face-neighbours is computed as

$$\epsilon_\rho = 2\,\max\!\left(\frac{|\rho_+ - \rho_0|}{|\rho_+ + \rho_0| + \rho_{\rm floor}},\; \frac{|\rho_0 - \rho_-|}{|\rho_0 + \rho_-| + \rho_{\rm floor}}\right). \tag{5}$$

If $\epsilon_\rho >$ `err_grad_d` along any axis, the cell is flagged.

(iv) **Pressure gradient ($\mathcal{P}$).** Analogous to the density gradient but applied to the thermal pressure:

$$\epsilon_P = 2\,\max\left(\frac{|P_+ - P_0|}{|P_+ + P_0| + P_{\rm floor}},\ \frac{|P_0 - P_-|}{|P_0 + P_-| + P_{\rm floor}}\right), \tag{6}$$

flagging when $\epsilon_P >$ `err_grad_p`.

(v) **Velocity gradient ($\mathcal{U}$).** A normalised velocity jump relative to the local sound speed:

$$\epsilon_v = 2\,\max\left(\frac{|v_+ - v_0|}{c_{s,+} + c_{s,0} + |v_+| + |v_0| + v_{\rm floor}},\ \frac{|v_0 - v_-|}{c_{s,0} + c_{s,-} + |v_0| + |v_-| + v_{\rm floor}}\right), \tag{7}$$

flagging when $\epsilon_v >$ `err_grad_u` in any component.

(vi) **Jeans length ($\mathcal{J}$).** When `jeans_refine`$(\ell) = N_J > 0$, the cell is flagged if the local Jeans length is shorter than N_J cell widths:

$$\lambda_{\rm J} = \sqrt{\frac{\pi\, c_s^2}{G\,\rho}} < N_J\,\Delta x_\ell, \tag{8}$$

implementing the Truelove et al. (1997) condition to prevent artificial fragmentation in self-gravitating gas. An additional density threshold `d_jeans_thre` restricts this criterion to cells with $n_{\rm H} \geq$ `d_jeans_thre`.

2.1.2 Geometry Mask (AND)

Ellipsoidal region ($\mathcal{G}$). When `r_refine`$(\ell) > -1$, only cells inside a generalised ellipsoidal region are permitted to refine. The test is

$$\left(\frac{2|\Delta x|}{r}\right)^{e_r} + \left(\frac{2|\Delta y|}{a\,r}\right)^{e_r} + \left(\frac{2|\Delta z|}{b\,r}\right)^{e_r} < 1, \tag{9}$$

where $r =$ `r_refine`, $a =$ `a_refine`, $b =$ `b_refine`, and $e_r =$ `exp_refine`. Cells outside this region are *un-flagged* regardless of physical criteria, restricting high resolution to a zoom-in target.

2.1.3 Cosmological Holdback (NOT)

Resolution holdback ($\mathcal{H}$). In cosmological runs with cooling, overly eager refinement at high redshift can exhaust memory before the epoch of interest. When `q_refine_holdback=.true.` and the physical cell size $\Delta x_\ell < 4^{1/N_{\rm dim}}\,\Delta x_{N_{\rm levelmax}}/a_{\rm exp}$ for levels $\ell >$ `nlevelmax_part` + 3, the *entire* level is blocked from further refinement:

$$\mathcal{H}(\ell) = \left[\ell > \ell_{\rm part} + \ell_{\rm collapse}\right] \wedge \left[\Delta x_\ell < 4^{1/N_{\rm dim}}\,\Delta x_{\rm min}/a\right], \tag{10}$$

where $\ell_{\rm part}$ is the particle initial-condition level (typically $\ell_{\rm min}$) and $\ell_{\rm collapse} = 3$. The first holdback level $\ell_{\rm hb}$ is the smallest ℓ satisfying equation (10):

$$\ell_{\rm hb} = \min\left\{\ell > \ell_{\rm part} + \ell_{\rm collapse}\ :\ \Delta x_\ell < 4^{1/N_{\rm dim}}\,\Delta x_{\rm min}/a\right\}. \tag{11}$$

2.2 Buffer Zones

After applying equation (3), flagged cells are expanded by `nexpand`(ℓ) cell widths in each direction via a dilation operator (`smooth_fine`). This cubic buffer ensures smooth refinement boundaries, preventing the Godunov solver from encountering abrupt level transitions.

3 RADIATIVE COOLING AND THE JEANS LENGTH

The cooling module is a critical component of the simulation. It determines the equilibrium temperature of the gas, which in turn controls the Jeans length $\lambda_{\rm J} \propto \sqrt{T/\rho}$ and hence the refinement decisions through the Jeans criterion $\mathcal{J}$. Errors in the cooling calculation — whether from the cooling rate tables, the UV background model, or the numerical integration scheme — propagate directly into errors in $\lambda_{\rm J}$ and may trigger (or suppress) refinement at the wrong density and temperature.

In this section we first analyse the limitations of the standard RAMSES cooling solver, then compare its underlying cooling rates against the Grackle equilibrium chemistry library (Smith et al. 2017), describe an alternative exact integration scheme based on the Townsend (2009) Y-function method, and finally present a multi-redshift table that tracks the evolving UV background across cosmic time.

3.1 The Standard RAMSES Cooling Solver

The RAMSES cooling module solves the energy equation

$$\frac{dT_2}{dt} = -\frac{2X_{\rm H}}{3k_{\rm B}}\,\frac{\Lambda_{\rm net}(n_{\rm H}, T_2)}{n_{\rm H}}, \tag{12}$$

where $T_2 \equiv T/\mu$ is the temperature per mean molecular weight and $\Lambda_{\rm net}$ is the net volumetric cooling rate (cooling − heating) in erg cm^{-3} s^{-1}.

The solver uses a semi-implicit Newton–Raphson iteration. At each sub-step, the temperature is updated as

$$T_2^{(n+1)} = T_2^{(n)}\,\frac{1 + \Lambda'/w_{\rm cool} - \Lambda/(T_2^{(n)}\,w_{\rm cool})}{1 + \Lambda'/w_{\rm cool}}, \tag{13}$$

where $\Lambda' \equiv d\Lambda_{\rm net}/dT_2$ and the adaptive step control parameter is $w_{\rm cool} = \max\big(|\Lambda|/(T_2\,\Delta_{\rm var}),\ w_{\rm max},\ -\Lambda'\,\Delta_{\rm var}\big)$ with $\Delta_{\rm var} = 4$.

The cooling rates $\Lambda_{\rm net}$ and their derivatives are interpolated from a pre-computed 2D table $\Lambda(n_{\rm H}, T_2)$, built at the start of the simulation for the current expansion factor a. The table contains separate contributions from:

- H/He collisional cooling (ionisation, recombination, bremsstrahlung, excitation) computed from collisional ionisation equilibrium (CIE; Sutherland & Dopita 1993);
- UV photo-heating from the Courty & Alimi (2004) background model;
- Compton cooling/heating off the CMB;
- metal cooling from the Cloudy 07 table (Plöckinger & Schaye 2015), scaled linearly with metallicity Z.

3.1.1 Overshooting Near Thermal Equilibrium

The Newton–Raphson scheme (equation 13) is designed for monotonic cooling (or heating). However, near the thermal equilibrium temperature $T_{\rm eq}$ where $\Lambda_{\rm net}(T_{\rm eq}) = 0$, the net rate changes sign and the update can overshoot. Specifically:

Sign reversal. Consider a cell at temperature $T > T_{\rm eq}$ that is cooling ($\Lambda_{\rm net} > 0$). A Newton–Raphson step decreases T_2. If the step is large enough to cross $T_{\rm eq}$, the cell enters the heating regime ($\Lambda_{\rm net} < 0$). On the next iteration, the solver increases T_2, potentially overshooting back above $T_{\rm eq}$. The result is an oscillation around $T_{\rm eq}$ that converges only after many sub-steps, or in the worst case does not converge within the 500-iteration limit.

Steep cooling function. The problem is exacerbated where the cooling function has a steep slope — in particular, near the hydrogen

Lyman-α peak at $T \sim 10^4$ K and the metal cooling peak at $T \sim 10^{4.5}$–$10^{5.5}$ K. In these temperature ranges, $|\Lambda'|$ is large and the linearised N-R step can dramatically overshoot. The adaptive step control $w_{\rm cool}$ mitigates this by limiting $\Delta T_2/T_2 \leq \Delta_{\rm var}$, but for cells undergoing rapid cooling (large $|\Lambda|/T_2$), the effective sub-step can still be too large to track the equilibrium accurately.

Cumulative error. Because T_2 enters the next hydrodynamic time-step as the updated thermal energy, any overshoot at step n feeds back into the pressure and hence the Jeans length at step n+1. A cell that overshoots to $T < T_{\rm eq}$ will have an artificially low Jeans length, potentially triggering spurious refinement. Conversely, a cell that undershoots (remains at $T > T_{\rm eq}$) may fail to trigger the Jeans criterion when it should.

Time-step coupling. The maximum sub-step size within `solve_cooling` is determined by the cooling time $t_{\rm cool} = (3/2)\, nk_{\rm B}T/|\Lambda_{\rm net}|$. When $\Lambda_{\rm net}$ is large, the solver takes many small sub-steps and is accurate. But near equilibrium, $|\Lambda_{\rm net}|$ is small, $t_{\rm cool}$ is large, and the solver takes fewer, larger steps — precisely where accuracy matters most for determining the equilibrium temperature.

3.1.2 Limitations of the 2D Table

Beyond the numerical integration, the cooling *rates* themselves have systematic biases:

(i) **No metallicity axis.** The RAMSES table is 2D: $(n_{\rm H}, T_2)$. Metal cooling is stored as a separate 1D function $\Lambda_Z(T)$ at solar metallicity, then scaled as $Z{\cdot}\Lambda_Z$. This linear scaling is correct for CIE but neglects the density-dependent ionisation corrections that become important at $n_{\rm H} > 1\,{\rm cm}^{-3}$, where photo-ionisation by the UV background cannot penetrate and the gas is self-shielded.

(ii) **CIE assumption.** The mean molecular weight μ is computed assuming collisional ionisation equilibrium at each $(n_{\rm H}, T)$ point. This neglects time-dependent ionisation (important during rapid cooling) and photo-ionisation at low temperatures (important in the diffuse IGM at $T < 10^4$ K), where the UV background maintains a residual ionisation fraction that CIE would predict to be negligible.

(iii) **UV background model.** The default Courty model is a polynomial fit calibrated against Cloudy 07. More recent UV background models (e.g. Haardt & Madau 2012; Faucher-Giguère 2020) differ significantly at $z > 3$, particularly in the He II reionisation epoch ($z \sim 3$–4) and at the earliest epochs where the UV intensity rises steeply.

(iv) **Temperature variable $T_2 = T/\mu$.** The solver works with T_2 rather than T. Since μ changes across the ionisation transition ($\mu \approx 1.22$ for neutral gas, $\mu \approx 0.59$ for fully ionised), a small change in T_2 maps to a factor-of-two change in physical temperature T across the transition. This amplifies interpolation errors near $T \sim 10^4$ K.

3.2 Comparison on the Density–Temperature Plane

To quantify the systematic differences, we compare the RAMSES net cooling rate against the Grackle equilibrium cooling library (Smith et al. 2017), which uses the Haardt & Madau (2012) UV background and includes density-dependent metal cooling with full non-LTE ionisation corrections. We compute $\Lambda_{\rm net}(n_{\rm H}, T)$ on a grid of 60×256 points spanning $n_{\rm H} = 10^{-8}$–$10^4\,{\rm cm}^{-3}$ and $T = 10$–10^9 K at $z = 5.1$ (the redshift of the HR5 test restart).

Low metallicity ($Z = 10^{-3}\,Z_\odot$). Fig. 1 compares the two models at the simulation average metallicity (`z_ave` $= 10^{-3}$). At this low metallicity, the net rate is dominated by H/He processes and UV photo-heating.

The lower-left panel shows the log ratio of the cooling rates, $\log_{10}(|\Lambda_{\rm RAMSES}|/|\Lambda_{\rm Grackle}|)$. Contour lines mark the thermal equilibrium ($\Lambda_{\rm net} = 0$) for each model: the yellow solid line for RAMSES and the green dashed line for Grackle. Three features are apparent:

(i) The *equilibrium temperature* differs by up to $\Delta \log T \sim 0.3$–0.5 at low densities ($n_{\rm H} \lesssim 10^{-2}\,{\rm cm}^{-3}$), reflecting the different UV backgrounds. Since $\lambda_{\rm J} \propto T^{1/2}$, this translates to a $\sim$ 40–80 per cent difference in the Jeans length at the same density.

(ii) At intermediate temperatures (10^4–10^5 K), the RAMSES rate is systematically *stronger* (redder in the ratio panel) by up to an order of magnitude. This arises because the CIE assumption overpredicts the neutral hydrogen fraction at these temperatures for low-density gas where the UV background maintains significant photo-ionisation.

(iii) At $T > 10^7$ K, both models converge to the bremsstrahlung limit $\Lambda \propto T^{1/2}\, n_{\rm H}^2$ and the ratio approaches unity, as expected.

The lower-right panel shows the mean molecular weight difference $\Delta\mu = \mu_{\rm RAMSES} - \mu_{\rm Grackle}$. The RAMSES CIE μ is systematically higher than the Grackle non-LTE value at $T > 10^5$ K. At intermediate temperatures ($T \sim 10^4$–10^5 K), the Grackle μ shows a strong *density dependence* due to the UV background maintaining partial ionisation at low densities — a feature entirely absent from the RAMSES model, where $\mu(T_2)$ is independent of density. This density-dependent μ has direct implications for the $T_2 \to T$ conversion and hence for the Jeans length evaluation.

Solar metallicity ($Z \approx Z_\odot$). Fig. 2 shows the same comparison at near-solar metallicity, where metal line cooling dominates the temperature range $T \sim 10^4$–$10^{5.5}$ K. The differences between the two models are significantly amplified:

- The Grackle table includes density-dependent ionisation corrections for metal species, leading to *up to* $\sim 10\times$ *stronger cooling* in the $T \sim 10^{4.5}$–10^5 K range compared to the CC07 CIE table. This is precisely the temperature range where infalling gas cools through the virial temperature of 10^{11}–$10^{12}\,{\rm M}_\odot$ haloes.
- The equilibrium temperature contours diverge more strongly, with the RAMSES equilibrium systematically higher at $n_{\rm H} < 10^{-1}\,{\rm cm}^{-3}$. This implies that the standard RAMSES solver overestimates the temperature of the warm-hot intergalactic medium (WHIM), reducing the Jeans-based refinement trigger and potentially missing structure in filaments.
- The UV suppression factor for metal cooling differs: the Courty model applies a redshift-dependent attenuation $f_{\rm UV}(n_{\rm H}, z)$ that damps metal cooling more strongly at low densities than the Grackle self-shielding prescription.

3.3 The Townsend Exact Integration Scheme

To address the overshooting problem of the N-R solver (Section 3.1.1), we implement the exact integration method of Townsend (2009), which transforms the cooling ODE into an analytically invertible form using the Y-function.

3.3.1 Y-Function Formulation

For a piecewise power-law cooling function $\Lambda(T) \propto T^{\alpha_j}$ in each temperature bin $[T_j, T_{j+1}]$, the integrated cooling variable is

$$Y(T) = \int_{T_{\rm ref}}^{T} \frac{dT'}{T'\,\hat{\Lambda}(T')} = \sum_k \Delta Y_k, \tag{14}$$

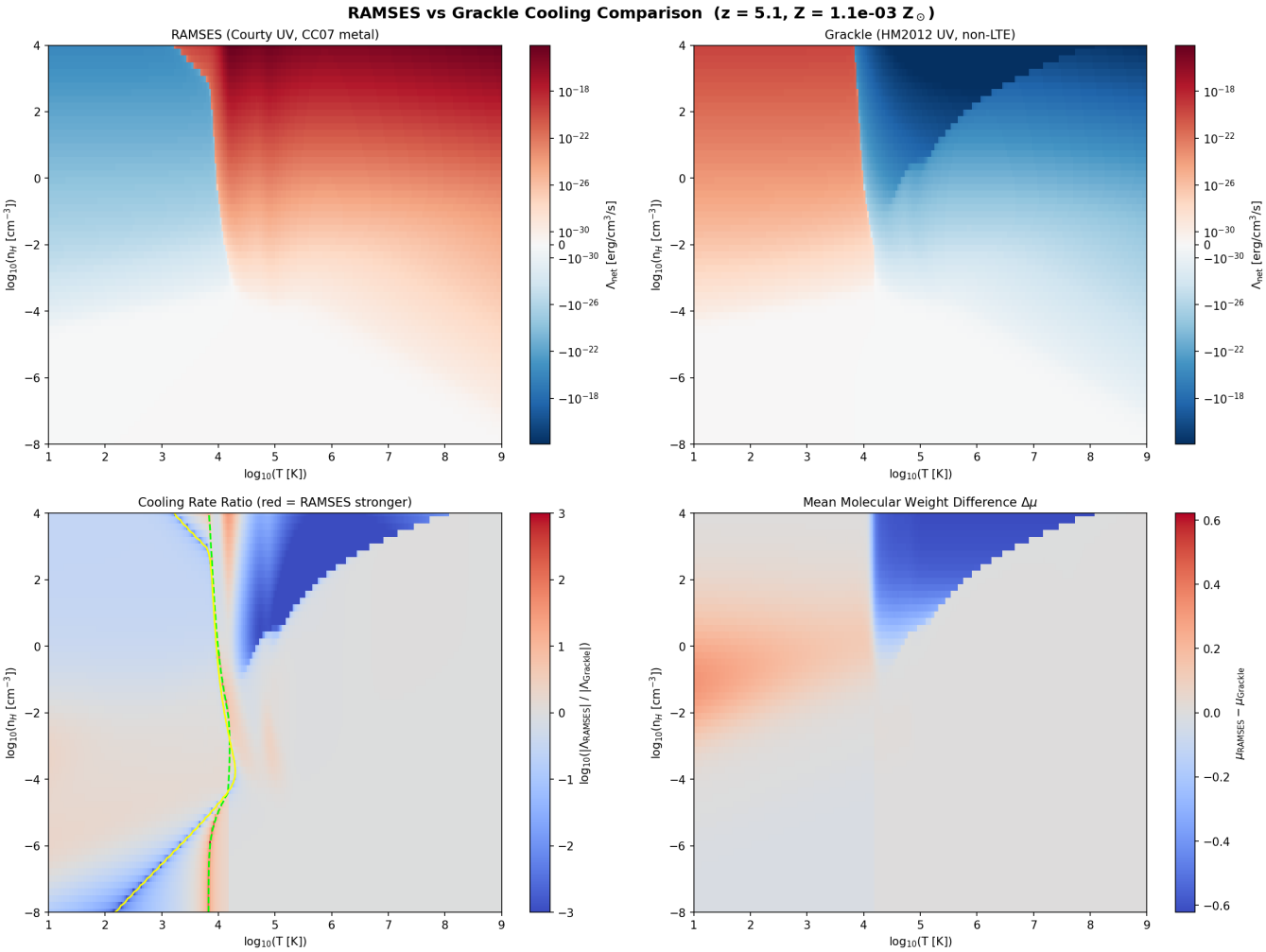

Figure 1. Comparison of the RAMSES and Grackle net cooling rates at $z = 5.1$ and $Z = 1.1 \times 10^{-3}\,Z_\odot$ (the simulation mean metallicity). *Top left:* RAMSES net cooling rate (Courty UV, Cloudy 07 CIE metals). *Top right:* Grackle net cooling rate (HM2012 UV, non-LTE equilibrium chemistry). Red shading indicates net cooling (energy loss); blue indicates net heating (energy gain). *Bottom left:* Log ratio of the absolute cooling rates, $\log_{10}(|\Lambda_{\rm RAMSES}|/|\Lambda_{\rm Grackle}|)$; yellow solid curve = RAMSES thermal equilibrium ($\Lambda = 0$), green dashed = Grackle equilibrium. Note the divergence of the equilibrium contours at $n_{\rm H} < 1\,{\rm cm}^{-3}$. *Bottom right:* Mean molecular weight difference $\mu_{\rm RAMSES} - \mu_{\rm Grackle}$, showing the density-dependent μ from Grackle's non-LTE treatment.

where $\hat{\Lambda} = \Lambda/\Lambda_{\rm ref}$ is the normalised cooling rate. Each segment contributes

$$\Delta Y_j = \begin{cases} \dfrac{T_j^{-\alpha_j} - T_{j+1}^{-\alpha_j}}{\alpha_j\,\hat{\Lambda}_j\,T_j^{-\alpha_j}} & (\alpha_j \neq 0), \\[2ex] \dfrac{\ln(T_j/T_{j+1})}{\hat{\Lambda}_j} & (\alpha_j = 0). \end{cases} \tag{15}$$

The exact temperature after a time-step Δt is obtained by inverting the relation

$$Y(T^{n+1}) = Y(T^n) + \Lambda_{\rm ref}\,\frac{n_{\rm H}}{E_0}\,\Delta t, \tag{16}$$

where $E_0 = (3/2)\,n_{\rm tot}\,k_{\rm B}$ is the thermal energy per unit temperature. The inversion is performed by binary search on the cumulative Y-table and local analytic inversion within the identified segment.

3.3.2 Advantages Over Newton–Raphson

The Y-function scheme has several fundamental advantages:

(i) **No overshooting.** The method integrates the cooling equation *exactly* for piecewise power-law rates. The temperature evolves monotonically towards the equilibrium; it cannot oscillate across $T_{\rm eq}$.

(ii) **No sub-stepping.** A single evaluation of equation (16) advances the solution by the full hydrodynamic time-step, regardless of $t_{\rm cool}/\Delta t$. The N-R solver requires up to 500 sub-steps when $\Delta t \gg t_{\rm cool}$.

(iii) **Correct equilibrium.** When $\Delta t \to \infty$, $Y \to Y_{\rm eq}$ and the cell relaxes exactly to $T_{\rm eq}$. The N-R solver only reaches $T_{\rm eq}$ asymptotically through iteration, and may settle on the wrong side if the sub-step structure interacts with the sign change.

(iv) **Deterministic.** The result depends only on $(T^n, n_{\rm H}, Z, \Delta t)$ and the cooling table, with no sensitivity to iteration count, convergence tolerance, or adaptive step heuristics.

The cost is a table lookup and a log/exp evaluation per cell (~few μs), comparable to a single N-R sub-step. Since the N-R solver typically requires 5–50 sub-steps, the exact scheme is faster for strongly cooling gas.

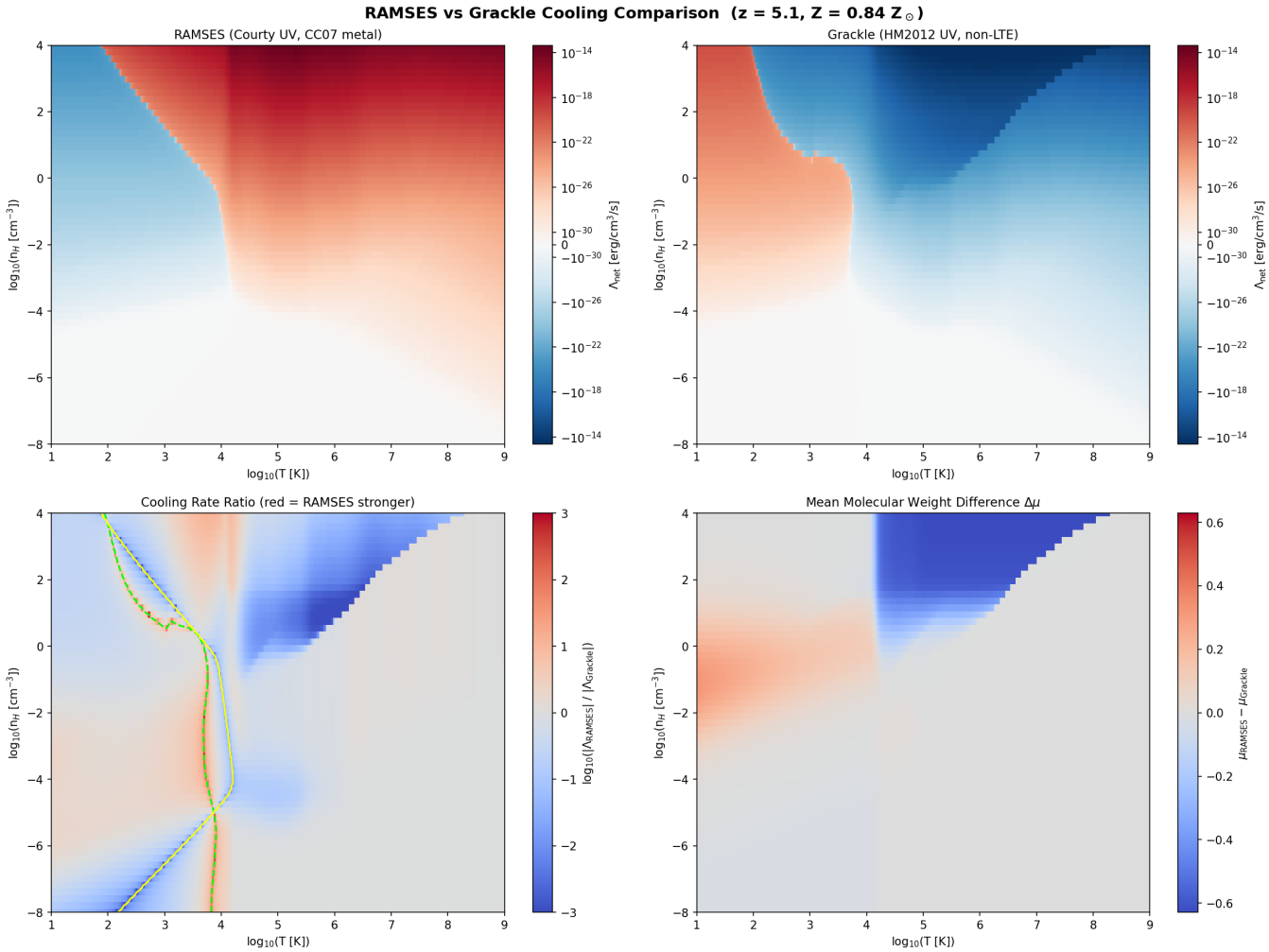

Figure 2. Same as Fig. 1 but at solar metallicity ($Z = 0.84\,Z_\odot$). Metal cooling dominates the net rate at $T \sim 10^4$–$10^{5.5}$ K, amplifying the model differences. Note the larger separation of the equilibrium contours (bottom-left) and the stronger $\Delta\mu$ pattern (bottom-right) compared to the low-metallicity case.

3.3.3 Implementation: Eunha Cooling Module

The exact integrator is implemented as a Fortran module (`eunha_cooling_mod.f90`) that reads a 3D Grackle equilibrium cooling table $\Lambda_{\rm net}(n_{\rm H}, Z, T)$ and pre-computes the cumulative Y-table. The module provides three runtime modes selected by the namelist parameter `cooling_method`:

- `'original'` — standard RAMSES N-R solver (default, no overhead);
- `'exact'` — Townsend exact integration replaces the N-R solver;
- `'compare'` — both solvers run; the N-R result is used for physics while per-step diagnostics report the difference $\Delta T_2^{\rm orig} - \Delta T_2^{\rm exact}$.

The Grackle table adds a third axis (metallicity Z) with 40 bins spanning 10^{-6}–$10\,Z_\odot$, compared to the 2D RAMSES table with linear Z scaling. The mean molecular weight $\mu(n_{\rm H}, Z, T)$ is also tabulated from Grackle's non-LTE equilibrium, replacing the RAMSES CIE calculation. The $T_2 \leftrightarrow T$ conversion uses an iterative scheme: $\mu \to T = T_2\,\mu \to \mu'(n_{\rm H}, Z, T) \to T' = T_2\,\mu'$, converging to < 0.1 per cent in 3 iterations.

3.4 Redshift-Dependent Cooling: Multi-z Tables

The cooling rate depends on redshift through the UV background intensity $J_\nu(z)$, which evolves as the ionising photon production rate rises during cosmic reionisation ($z \gtrsim 6$), peaks near $z \sim 2$–3, and declines towards $z = 0$ as the star formation rate density decreases (Haardt & Madau 2012; Faucher-Giguère 2020). The equilibrium temperature $T_{\rm eq}(n_{\rm H}, z)$ therefore evolves with redshift even at fixed density.

3.4.1 How the standard solver tracks redshift

The standard RAMSES cooling module calls `set_table(aexp)` at each coarse time-step, rebuilding the 2D cooling table $\Lambda(n_{\rm H}, T_2)$ using the Courty UV background model evaluated at the current expansion factor a. This analytic UV model is a polynomial fit to Cloudy 07, so the table rebuild is inexpensive and correctly tracks the redshift evolution of the background — within the limitations of the Courty model itself (Section 3.1.2).

3.4.2 *The single-z limitation of the Grackle table*

The Grackle-based Eunha module (`eunha_cooling_mod.f90`) initially loaded a single pre-computed 3D table at a fixed redshift (e.g. $z = 5.1$ for the HR5 restart). While the cooling rates are more accurate than the RAMSES 2D table at that redshift, the table becomes increasingly wrong as the simulation evolves away from the tabulated z. In particular:

- At $z < z_{\rm table}$, the UV background is stronger (more photoheating), so the true $T_{\rm eq}$ is higher than the tabulated value. The exact integrator would drive cells to an *artificially low* equilibrium temperature, underestimating $\lambda_{\rm J}$ and potentially triggering spurious refinement.
- At $z > z_{\rm table}$, the UV background is weaker, and the exact integrator would settle on a $T_{\rm eq}$ that is too high, overestimating $\lambda_{\rm J}$ and suppressing refinement.
- The most rapid UV evolution occurs during hydrogen reionisation ($z \sim 6$–8) and He II reionisation ($z \sim 3$–4), where $T_{\rm eq}$ can change by factors of 2–3 over $\Delta z \sim 1$.

3.4.3 *Multi-z table and runtime interpolation*

We address this by pre-computing Grackle equilibrium tables at multiple redshifts and interpolating at runtime. The offline generator (`generate_multi_z_table.py`) calls the Grackle library at each redshift snapshot to produce a single multi-z binary file containing N_z copies of the 3D table $(n_{\rm H}, Z, T)$.

The default sampling uses 10 redshift snapshots:

$$z \in \{0.0,\ 0.5,\ 1.0,\ 1.5,\ 2.0,\ 3.0,\ 4.0,\ 5.0,\ 7.0,\ 10.0\}, \tag{17}$$

with denser spacing at low z ($\Delta z = 0.5$ for $z \leq 2$) where the UV background evolves most rapidly relative to the Hubble time, and coarser spacing at high z ($\Delta z \geq 1$).

At each coarse time-step, `eunha_interp_redshift(z)` performs the following:

(i) Finds the bracketing snapshots $z_i \leq z < z_{i+1}$ and computes the linear interpolation weight $w = (z - z_i)/(z_{i+1} - z_i)$.

(ii) Interpolates the net cooling rate and mean molecular weight:

$$\Lambda_{\rm net}(n_{\rm H}, Z, T; z) = (1-w)\,\Lambda^{(i)} + w\,\Lambda^{(i+1)}, \tag{18}$$

$$\mu(n_{\rm H}, Z, T; z) = (1-w)\,\mu^{(i)} + w\,\mu^{(i+1)}. \tag{19}$$

(iii) Rebuilds the cumulative Y-table for all $N_{\rm dens} \times N_{\rm metal} = 60 \times 40 = 2400$ cells (OpenMP parallelised).

(iv) Caches the current redshift; the rebuild is skipped if $|\Delta z| < 0.01$.

The total file size for 10 snapshots on the standard $60 \times 40 \times 256$ grid is $10 \times 2 \times 614400 \times 8 \approx 94$ MB, which is distributed with the source code. Backward compatibility with single-z files is maintained through automatic format detection (the first `int32` in a multi-z file encodes N_z, which is distinguishable from the grid dimension $L = 60$).

3.4.4 *Impact on cooling overshoot*

The multi-z table eliminates a systematic error that compounds with the numerical overshoot discussed in Section 3.1.1. Consider a simulation at $z = 2$ using a table computed at $z = 5$:

(i) The UV background at $z = 2$ is ~ 3–5× stronger than at $z = 5$ (Haardt & Madau 2012), so $T_{\rm eq}^{z=2} > T_{\rm eq}^{z=5}$.

(ii) A cell cooling towards equilibrium at $z = 2$ would be driven to the *wrong* (lower) $T_{\rm eq}^{z=5}$ by the stale table.

(iii) Even with the exact integrator (no numerical overshoot), the cell reaches an incorrect equilibrium, yielding a systematically too-small Jeans length.

(iv) The N-R solver compounds this: it may overshoot below the already-too-low $T_{\rm eq}^{z=5}$, further reducing $\lambda_{\rm J}$.

With the multi-z table, the Eunha module tracks the correct $T_{\rm eq}(z)$ at every coarse step, matching the RAMSES solver's `set_table(aexp)` behaviour but with the improved Grackle rates and the exact integration scheme. The combination eliminates both the systematic error (wrong UV background) and the numerical error (N-R overshoot), providing the most accurate equilibrium temperature determination available within the operator-split cooling framework.

3.5 Implications for the Jeans Criterion

The combined effect of the cooling rate differences (Figs 1–2), the integration scheme, and the redshift evolution of the UV background impacts the Jeans refinement criterion through four channels:

1. Equilibrium temperature. The Jeans length at thermal equilibrium is $\lambda_{\rm J}^{\rm eq} \propto (T_{\rm eq}/\rho)^{1/2}$. Since $T_{\rm eq}^{\rm RAMSES} \neq T_{\rm eq}^{\rm Grackle}$ (bottom-left panels), the two solvers will flag different cells. At low densities, the RAMSES equilibrium is systematically *higher* ($T_{\rm eq} \sim 2$–3× warmer), yielding a larger Jeans length and less refinement. This means the standard solver may *underestimate* the refinement needed in filaments and the circumgalactic medium.

2. Cooling overshoot near $T_{\rm eq}$. The N-R solver's tendency to overshoot $T_{\rm eq}$ (Section 3.1.1) creates transient temperature errors that affect the *instantaneous* Jeans length even if the time-averaged temperature is correct. A cell that overshoots to $T < T_{\rm eq}$ will have a transiently small $\lambda_{\rm J}$, triggering refinement that may not be needed. Conversely, a cell that oscillates above $T_{\rm eq}$ will have a transiently large $\lambda_{\rm J}$, suppressing refinement that *is* needed. The exact integrator eliminates this noise entirely.

3. Mean molecular weight. The $T_2 = T/\mu$ representation amplifies errors in μ. With the RAMSES CIE model, μ is a function of T_2 alone (density-independent). The Grackle non-LTE μ depends on both $n_{\rm H}$ and T, with $|\Delta\mu| \sim 0.3$–0.6 in the transition zone ($T \sim 10^4$–10^5 K; bottom-right panels). Since $T = T_2\,\mu$, a 50 per cent error in μ produces a 50 per cent error in T and hence a ~ 22 per cent error in $\lambda_{\rm J}$, comparable to the factor-of-two ratio that separates adjacent refinement levels.

4. Redshift evolution of $T_{\rm eq}$. The thermal equilibrium temperature evolves with redshift as the UV background strengthens (towards $z \sim 2$) and then weakens (towards $z = 0$). A fixed-z Grackle table ignores this evolution, producing a systematic offset in $T_{\rm eq}$ that grows with $|z - z_{\rm table}|$. At the epoch of maximum UV background ($z \sim 2$–3), this offset can reach factors of 2–3 in $T_{\rm eq}$ if the table was computed at $z = 5$, corresponding to a ~ 40–70 per cent error in $\lambda_{\rm J}$. The multi-z interpolation (Section 3.4) eliminates this systematic bias, ensuring that $T_{\rm eq}$ tracks the correct UV background at every epoch.

Table 1 summarises the systematic differences and their impact on the Jeans length.

4 THE STARBURST PROBLEM

The holdback mechanism (Section 2) blocks *all* refinement criteria at levels $\ell \geq \ell_{\rm hb}$, including the Jeans criterion. This creates two related problems:

Table 1. Systematic differences between the RAMSES and Grackle/Townsend cooling implementations and their impact on the Jeans length.

Component	RAMSES	Grackle/Townsend	λ_J impact
Integration	N-R implicit (sub-stepping)	Exact Y-function (single step)	Removes overshoot noise
UV background	Courty (Cloudy 07)	Haardt & Madau (2012) (Cloudy 13)	T_{eq} shift → 40–80 per cent
Metal cooling	CC07 CIE, linear Z scaling	3D non-LTE, density-dep. Z	Up to 10× at $10^{4.5}$ K
μ model	CIE, density-independent	Non-LTE, $\mu(n_H, Z, T)$	$\lvert\Delta\mu\rvert \sim 0.3$–$0.6$
Table dimension	2D: (n_H, T_2)	3D: (n_H, Z, T)	Metallicity resolution
Redshift	Rebuilt per step (`set_table`)	Multi-z interpolation (10 snapshots)	Tracks $T_{eq}(z)$ correctly

Artificial fragmentation. The Jeans length criterion $\mathcal{J}$ differs fundamentally from the other refinement triggers: it is a *numerical stability* requirement (Truelove et al. 1997), not merely a resolution preference. Violating the Jeans condition leads to artificial fragmentation — a qualitatively wrong result — regardless of epoch. Yet in the standard RAMSES logic (equation 3), the cosmological holdback $\mathcal{H}$ blocks $\mathcal{J}$ uniformly with all other criteria. At high redshift, dense self-gravitating regions may therefore be held at insufficient resolution, producing spurious fragmentation that propagates into lower-redshift structure.

Abrupt SFR transition. When the holdback releases level ℓ (i.e. $\mathcal{H}(\ell)$ transitions from true to false), all accumulated criteria fire simultaneously. Many cells that have been gravitationally collapsing during the holdback period now suddenly satisfy the refinement condition, leading to a burst of new cells and an abrupt spike in the star formation rate.

5 JEANS BYPASS: DESIGN AND IMPLEMENTATION

5.1 Modified Refinement Logic

The key observation is that the Jeans length criterion is *cell-local*: it depends only on the density and temperature of the cell itself (`uold`), with no neighbour information required. This allows a lightweight evaluation even when the full refinement machinery (which requires neighbour gathering for $\mathcal{D}, \mathcal{P}, \mathcal{U}$) is skipped by the holdback.

With `jeans_bypass_holdback=.true.`, the refinement logic becomes

$$\mathcal{R}' = \Big[\neg\mathcal{H} \wedge \mathcal{G} \wedge (\mathcal{M}\vee\mathcal{S}\vee\mathcal{D}\vee\mathcal{P}\vee\mathcal{U}\vee\mathcal{J})\Big] \vee \Big[\mathcal{H} \wedge \mathcal{J} \wedge (\ell < \ell_{hb}+n_{bypass})\Big] \tag{20}$$

where $n_{bypass} \equiv$ `n_jeans_bypass_levels` (default 1). The geometry mask $\mathcal{G}$ still applies through the non-holdback branch, preserving zoom-in spatial restrictions. Only the temporal holdback $\mathcal{H}$ is bypassed for $\mathcal{J}$, and only within a limited number of levels.

This modification is analogous to the existing treatment of sink particles: $\mathcal{S}$ is already evaluated inside the holdback-protected block, but unlike $\mathcal{J}$ it only triggers in cells that already host a sink. Extending the same exemption to $\mathcal{J}$ ensures numerical integrity for gas that is *about to* form a sink.

5.2 Level Limitation: Preventing Cascades

Without a level limit ($n_{bypass} \to \infty$), the Jeans bypass creates a catastrophic cascade. When a cell at the holdback level ℓ_{hb} is flagged by the Jeans criterion, child cells are created at level $\ell_{hb} + 1$. Since level $\ell_{hb} + 1$ is also in the holdback zone, the bypass applies again, flagging cells at $\ell_{hb} + 2$, and so on until ℓ_{max}. This rapidly exhausts available memory.

Table 2. Cascade refinement without level limit ($a = 0.164$, $\ell_{hb} = 14$, $\ell_{max} = 17$). All levels ≥ 14 are in the holdback zone.

Level	$\Delta x/\Delta x_{min}$	Holdback	Bypass	$\mathcal{J}$ fires	Result
≤ 12	≥ 32	No	—	Yes (normal)	Normal
13	16	No	—	Yes (normal)	Normal
14	8	Yes	Active	Yes	→ lv. 15 created
15	4	Yes	Active	Yes	→ lv. 16 created
16	2	Yes	Active	Yes	→ lv. 17 created
17	1	ℓ_{max}	—	—	OOM crash

The cascade occurs because the Jeans length in radiatively cooling gas (below the star-formation density threshold) shrinks without bound as the temperature drops. There is no physical mechanism to halt the Jeans criterion from triggering at every successive level — only the polytropic temperature floor (Section 6) or a hard level limit can stop it.

Table 2 shows the progression observed in a test run without level limit.

The memory footprint grew from 81 per cent to 88.5 per cent in a single coarse step before crashing. The fine-step time-step also decreased by 8× due to subcycling at levels 15–17.

With $n_{bypass} = 1$, only level $\ell_{hb} = 14$ is bypassed, creating cells at level 15 but no further. This provides a controlled pre-refinement before the holdback releases.

6 ROLE OF THE POLYTROPIC TEMPERATURE FLOOR

6.1 Jeans Length in the Polytropic Regime

In RAMSES, the star-formation module enforces a polytropic equation of state above the density threshold $n_\star$:

$$T(n) = T_{2,\star}\left(\frac{n}{n_\star}\right)^{g_\star-1}, \qquad n \geq n_\star, \tag{21}$$

with default parameters $T_{2,\star} = 750\,\mathrm{K}$, $n_\star = 0.1\,\mathrm{cm}^{-3}$, $g_\star = 5/3$. This floor is applied in the cooling routine and is already reflected in `uold`, so the Jeans length computation in `jeans_length_refine()` automatically includes the polytropic stabilisation.

Substituting equation (21) into equation (8):

$$\lambda_J \propto \sqrt{\frac{T}{\rho}} \propto \sqrt{\rho^{g_\star-1}/\rho} = \rho^{(g_\star-2)/2}. \tag{22}$$

For $g_\star = 5/3$: $\lambda_J \propto \rho^{-1/6}$. The Jeans length *decreases very slowly* with density — one or two additional levels of refinement are typically sufficient to resolve it.

In contrast, for gas below $n_\star$ (no polytropic floor), radiative cooling can drive T to very low values while ρ remains moderate. This yields $\lambda_J \ll \Delta x_\ell$ at many levels, causing the cascade.

Table 3. Two complementary mechanisms for preventing the bypass cascade.

Mechanism	Type	Behaviour
`n_jeans_bypass_levels`	Hard limit	Caps extra levels regardless of physics
`d_jeans_thre` $\approx n_\star$	Physics-based	Polytropic floor limits λ_J decrease

6.2 Self-regulation via `d_jeans_thre`

The density threshold parameter `d_jeans_thre` (in units of n_H [cm^{-3}]) restricts the Jeans criterion to cells above a given density:

$$\mathcal{J}(\text{cell}) = \begin{cases} N_J\,\Delta x_\ell \geq \lambda_J & \text{if } n \geq \texttt{d_jeans_thre}, \\ \text{false} & \text{otherwise.} \end{cases} \tag{23}$$

Setting `d_jeans_thre` $\approx n_\star$ restricts the Jeans criterion to the polytropic regime. The cascade then self-limits through the following mechanism:

(i) At level $\ell_{\rm hb}$: only cells with $n \geq n_\star$ are checked. The polytropic floor guarantees $T \geq T_{2,\star}$, so λ_J is relatively large.
(ii) If $N_J\,\Delta x_{\ell_{\rm hb}} > \lambda_J$, the cell is flagged and children at $\ell_{\rm hb}+1$ are created.
(iii) At level $\ell_{\rm hb}+1$: Δx is halved, density is approximately unchanged, λ_J is approximately unchanged.
(iv) $N_J\,\Delta x_{\ell_{\rm hb}+1} = N_J\,\Delta x_{\ell_{\rm hb}}/2 < \lambda_J \Rightarrow$ Jeans criterion is *satisfied* $\Rightarrow$ no further flagging $\Rightarrow$ cascade stops naturally.

This provides a *physics-based* self-regulation, complementary to the *hard limit* imposed by `n_jeans_bypass_levels`.

When `d_jeans_thre` = 0 (default), *all* cells are checked, including low-density gas where cooling produces arbitrarily small λ_J. This is the root cause of the unlimited cascade observed in Section 5.2.

7 IMPLEMENTATION DETAILS

7.1 Code Structure

The implementation modifies two source files (plus parameter declaration and namelist reading).

`flag_utils.kjhan.f90` — `userflag_fine()`. The holdback early-return is modified from a simple `return` to a conditional block:

```
if(prevent_refine .and. q_refine_holdback) then
   if(jeans_bypass_holdback .and. hydro &
      .and. n_jeans_bypass_levels > 0) then
      ! Find the first holdback level
      dx_min = (0.5D0**nlevelmax)*scale
      holdback_threshold = &
        (4d0**(1d0/ndim))*(dx_min/aexp)
      holdback_first = nlevelmax + 1
      do jlevel = nlevelmax_part &
         + nlevel_collapse+1, nlevelmax
         dx_test = (0.5d0**jlevel)*scale
         if(dx_test < holdback_threshold) then
            holdback_first = jlevel
            exit
         endif
      end do
      ! Only bypass within allowed range
      if(ilevel < holdback_first &
         + n_jeans_bypass_levels) then
         call jeans_only_flag(ilevel)
      endif
   endif
   return
endif
```

`hydro_flag.kjhan.f90` — `jeans_only_flag()`. A new lightweight subroutine that:

(i) Loops over active grids with OpenMP parallelism.
(ii) Computes cell indices without neighbour gathering.
(iii) Calls the existing `jeans_length_refine()` subroutine (no code duplication).
(iv) Sets `flag1` for flagged cells and updates `nflag`.

This is possible because `jeans_length_refine()` only accesses cell-local data (`uold(ind_cell,...)`), unlike the gradient-based criteria $\mathcal{D}, \mathcal{P}, \mathcal{U}$.

7.2 Safety Properties

- **Default off**: With `jeans_bypass_holdback=.false.` (default), the new code path is never entered. Bit-identical to the unmodified code.
- **No neighbour access**: `jeans_only_flag()` is safe to call at any point in the refinement cycle.
- **OMP safety**: `flag1` writes are grid-partitioned; `nflag` uses reduction.
- **Hard level limit**: `n_jeans_bypass_levels` provides a fail-safe even if `d_jeans_thre` is misconfigured.

7.3 Summary of Parameters

Table 4 lists all refinement-related namelist parameters, including the new bypass parameters.

The recommended configuration for cosmological runs with star formation is:

```
&REFINE_PARAMS
m_refine          = 9*8.,
jeans_refine      = 9*4.,
jeans_bypass_holdback  = .true.
n_jeans_bypass_levels  = 1
d_jeans_thre      = 0.1      ! = n_star
q_refine_holdback = .true.
/
```

8 TEST RESULTS

All tests use the HR5 medium-box setup: $\ell_{\rm min} = 9$, $\ell_{\rm max} = 17$, 12 MPI ranks, OMP=5, restarting from output 5 ($a = 0.164$, $z \approx 5.1$, nstep_coarse=241).

8.1 Bit-identical Verification

With `jeans_bypass_holdback=.false.` (default) and no `jeans_refine`, the new binary produces results identical to the reference run: econs=6.13×10^{-4} at step 242, econs=6.71×10^{-4} at step 243.

8.2 Unlimited Bypass

With `jeans_refine=9*4.`, `jeans_bypass_holdback=.true.`, `d_jeans_thre=0`, and no level limit: cascade refinement from level 14 through 17 within a single coarse step. Memory grew from 81 per cent to 88.5 per cent before OOM crash. Fine-step time-step decreased by 8× due to subcycling.

8.3 Limited Bypass ($n_{\rm bypass} = 1$)

With `n_jeans_bypass_levels=1`: only level 14 is bypassed, cells at level 15 may be created but no further cascade.

Table 4. AMR refinement parameters in the `&AMR_PARAMS` and `&REFINE_PARAMS` namelists. New parameters are marked with ⋆.

Parameter	Default	Role
`m_refine`	8.0	Lagrangian mass threshold (per level)
`sink_refine`	F	Force max level around sinks
`err_grad_d`	−1	Density gradient threshold
`err_grad_p`	−1	Pressure gradient threshold
`err_grad_u`	−1	Velocity gradient threshold
`jeans_refine`	−1	Jeans length in cells (per level)
`d_jeans_thre`	0	Density threshold for Jeans ($n_{\rm H}$ cm^{-3})
`r_refine`	−1	Geometry region diameter (per level)
`a_refine`	1.0	Ellipticity Y/X
`b_refine`	1.0	Ellipticity Z/X
`exp_refine`	2.0	Norm exponent
`x/y/z_refine`	0.5	Region centre
`nexpand`	1	Buffer zone width (per level)
`q_refine_holdback`	F	Cosmological holdback
⋆ `jeans_bypass_holdback`	F	Enable Jeans bypass of holdback
⋆ `n_jeans_bypass_levels`	1	Max extra levels beyond $\ell_{\rm hb}$

8.4 Future: `d_jeans_thre` = $n_\star$

A test with `d_jeans_thre=0.1` will verify the physics-based self-regulation, potentially allowing $n_{\rm bypass} > 1$ without cascade, since only gas in the polytropic regime would trigger the Jeans criterion.

9 DISCUSSION AND FUTURE WORK

9.1 Physical Motivation

The holdback mechanism was designed to prevent over-refinement from the mass criterion at early times. However, it inadvertently blocks the Jeans criterion, which has a different physical motivation: resolving the Jeans length prevents artificial fragmentation and ensures numerically converged star formation (Truelove et al. 1997; Federrath et al. 2011).

By allowing the Jeans criterion to operate ahead of the holdback release, we expect:

- Dense gas that will form stars is pre-refined before the holdback releases.
- When the holdback releases, the transition is smoother because the densest regions are already at the appropriate resolution.
- The SFR curve shows a gradual onset rather than an abrupt step.

9.2 Interplay of the Three Parameters

The bypass is controlled by three parameters that form a layered defence:

Layer 1: `jeans_bypass_holdback` (on/off). Master switch. When off (default), the system is identical to standard RAMSES.

Layer 2: `d_jeans_thre` (physics-based regulation). Restricts the Jeans check to dense gas where the polytropic floor operates. In this regime, $\lambda_{\rm J} \propto \rho^{-1/6}$ varies slowly, and 1–2 extra refinement levels suffice to resolve it. Provides self-consistent, physics-driven cascade termination.

Layer 3: `n_jeans_bypass_levels` (hard safety limit). Absolute cap on the number of extra levels. Prevents OOM regardless of physics configuration. Essential when `d_jeans_thre=0` (all cells checked).

9.3 Problems with Additional Temperature and Pressure Floors

A common approach to enforcing the Jeans criterion in cosmological simulations is to introduce an *additional* temperature or pressure floor beyond the star-formation polytropic floor (equation 21). Several major simulation projects adopt a resolution-dependent pressure floor of the form

$$P_{\rm floor} = \frac{G}{\pi\,\gamma}\,N_J^2\,\rho\,(\Delta x_\ell)^2, \tag{24}$$

which ensures $\lambda_{\rm J} \geq N_J\,\Delta x_\ell$ at all times (e.g. Schaye et al. 2015; Hopkins et al. 2018). Equivalently, this corresponds to a density-dependent temperature floor $T_{\rm floor} \propto \rho\,(\Delta x_\ell)^2$. While this *guarantees* that the Jeans criterion is never violated, it introduces several systematic problems:

(i) Artificial suppression of fragmentation. The pressure floor prevents gas from cooling below $T_{\rm floor}$, even when the physical equilibrium temperature $T_{\rm eq}$ is much lower. In the density range $n_\star < n < n_{\rm floor}$ (where $n_{\rm floor}$ is the density at which $T_{\rm floor}$ exceeds the polytropic floor), the gas is held at an artificially high temperature. This suppresses the thermal instability that drives the formation of the cold, dense phase of the ISM (Sutherland & Dopita 1993; Wiersma, Schaye & Smith 2009), potentially preventing the formation of molecular clouds and reducing the star formation rate.

(ii) Resolution-dependent thermodynamics. Since $P_{\rm floor} \propto (\Delta x_\ell)^2$, the effective equation of state of the gas depends on the grid resolution. At higher resolution ($\ell_{\rm max}$ increased), Δx decreases and the floor drops, allowing gas to cool further and fragment into smaller structures. The resulting star formation rate, stellar mass function, and galaxy morphologies are therefore resolution-dependent — not converging to a physical answer as resolution increases, but rather revealing new fragmentation at each resolution level.

(iii) Conflict with the polytropic floor. The star-formation polytropic floor (equation 21) is designed to model the unresolved physics of the multiphase ISM above $n_\star$: it represents the effective equation of state of a turbulent, star-forming medium (Schaye et al. 2015). An additional Jeans pressure floor that is active *below* $n_\star$ creates a discontinuity in the effective equation of state at $n = n_\star$ if the two floors do not match. More fundamentally, the Jeans floor operates on *numerical* grounds (preventing artificial fragmentation) while the polytropic floor operates on *physical* grounds (modelling the unresolved ISM), and conflating the two obscures the distinction between numerical and physical effects.

(iv) Masking of cooling errors. The pressure floor removes the consequences of cooling inaccuracies (Section 3.1.1) by overriding the computed temperature whenever it falls below $T_{\rm floor}$. While this prevents the most extreme manifestation of cooling overshoot (spurious fragmentation), it also prevents the diagnosis of the underlying problem. A cell that overshoots to $T \ll T_{\rm eq}$ due to the N-R integration error is silently corrected by the floor, but the energy error remains in the hydrodynamic variables. Over many time-steps, these hidden energy errors can accumulate and affect the large-scale thermodynamic state.

(v) Incompatibility with turbulent Jeans analysis. If the Jeans criterion is to be extended to include turbulent pressure support (equation 2), the floor must also be modified to account for $\sigma_{\rm turb}$. But the turbulent velocity dispersion varies in space and time, making a simple analytic floor impossible. The approach advocated in this paper — *resolving* the Jeans length through AMR rather than *imposing* it through a floor — is naturally compatible with a turbulence-dependent criterion, since the refinement condition (equation 8) can be straightforwardly generalised to use $c_{\rm eff}^2 = c_s^2 + \sigma_{\rm turb}^2$ without modifying any thermodynamic floor.

In summary, while a pressure floor provides a brute-force guarantee that the Jeans criterion is satisfied, it does so at the cost of altering the thermodynamic state of the gas. Our approach — combining accurate cooling (Section 3) with Jeans-aware refinement (Section 5) — achieves the same numerical stability without modifying the gas physics.

9.4 Turbulence and the Effective Jeans Length

The analysis presented in this paper treats the Jeans length as a purely thermal quantity (equation 8). In reality, turbulent motions in the ISM provide additional pressure support that modifies the effective stability criterion. Following Chandrasekhar (1951) and Bonazzola et al. (1987), the turbulent Jeans length is

$$\lambda_{\rm J}^{\rm turb} = \sqrt{\frac{\pi\,(c_s^2 + \sigma_{\rm turb}^2)}{G\,\rho}}, \tag{25}$$

where $\sigma_{\rm turb}$ is the turbulent velocity dispersion on the grid scale. In the limit $\sigma_{\rm turb} \gg c_s$, the turbulent Jeans length can exceed the thermal Jeans length by an order of magnitude, fundamentally changing the refinement pattern.

Subgrid-scale turbulence models (Schmidt, Niemeyer & Hillebrandt 2006; Schmidt & Federrath 2011) evolve a turbulent energy density $e_{\rm turb}$ alongside the hydrodynamic variables, providing a local estimate of $\sigma_{\rm turb} = (2e_{\rm turb}/\rho)^{1/2}$. Rennehan (2021) applied this approach to cosmological simulations, showing that SGS turbulence can suppress artificial fragmentation in the circumgalactic medium. Semenov, Kravtsov & Gnedin (2016) demonstrated that a turbulence-regulated star formation prescription produces more realistic star formation histories than a simple density-threshold model.

In galactic discs, turbulence is driven by a combination of supernova feedback, gravitational instabilities, magnetorotational instability, and large-scale shear (Robertson & Kravtsov 2008; Kim & Ostriker 2017; Hopkins et al. 2018). The relative importance of these drivers varies with environment: supernova feedback dominates in star-forming regions, while gravitational infall and accretion shocks drive turbulence in the CGM and IGM (Iapichino et al. 2011).

Incorporating equation (25) into the AMR refinement criterion requires:

(i) A local estimator of $\sigma_{\rm turb}$ — either from resolved velocity gradients (Federrath & Klessen 2012) or from a dedicated SGS model (Schmidt, Niemeyer & Hillebrandt 2006; Rennehan 2021).

(ii) Modification of `jeans_length_refine()` to use the effective sound speed $c_{\rm eff}^2 = c_s^2 + \sigma_{\rm turb}^2$ in place of c_s^2.

(iii) Re-evaluation of the cascade analysis (Section 5.2), since turbulent pressure *increases* $\lambda_{\rm J}^{\rm turb}$ and may naturally prevent the cascade without a hard level limit.

This turbulence-dependent Jeans criterion is a natural extension of the framework presented here, and will be the subject of a future paper.

Table 5. Comparison of cascade prevention by turbulence and holdback. The turbulent Mach numbers are representative values for each environment.

Environment	$\mathcal{M}_{\rm turb}$	$\Delta\ell_{\rm turb}$	$\Delta\ell_{\rm hb}(z{=}5)$	Sufficient?
IGM ($z \sim 5$)	0.1–0.5	0.0–0.2	3.8	No
Filaments	1–3	0.5–1.7	3.8	No
Halo virial region	1–3	0.5–1.7	3.8	No
Cold streams	3–10	1.7–3.3	3.8	Marginal
ISM (cold, dense)	5–10	2.3–3.3	—[a]	N/A

[a] Polytropic floor already prevents cascade in this regime.

9.5 Can Turbulence Replace the Holdback?

A natural question is whether turbulent pressure support alone can prevent the refinement cascades that the holdback mechanism was designed to suppress, thereby rendering the holdback unnecessary. We present both a quantitative scaling analysis and a direct numerical test.

Scaling analysis. Including turbulence, the Jeans length increases by a factor $\lambda_{\rm J}^{\rm turb}/\lambda_{\rm J} = (1 + \mathcal{M}_{\rm turb}^2)^{1/2}$, where $\mathcal{M}_{\rm turb} = \sigma_{\rm turb}/c_s$ is the turbulent Mach number. In terms of refinement levels, turbulence prevents the cascade from extending by

$$\Delta\ell_{\rm turb} = \tfrac{1}{2}\log_2(1 + \mathcal{M}_{\rm turb}^2) \tag{26}$$

levels. Table 5 compares this with the holdback, which prevents approximately $\Delta\ell_{\rm hb} \approx \log_2(4^{1/3}/a_{\rm exp})$ levels at a given expansion factor.

At $z = 5$ ($a_{\rm exp} = 1/6$), the holdback suppresses ~3.8 levels of refinement. Turbulence can match this only for $\mathcal{M}_{\rm turb} \gtrsim 10$, which occurs exclusively in the cold, dense ISM — where the polytropic floor already prevents the cascade. In the diffuse IGM and filaments where the holdback is most critical, typical Mach numbers are $\mathcal{M}_{\rm turb} \sim 0.1$–3, providing at most 1–2 levels of protection. Furthermore, the resolved velocity dispersion scales as $\sigma_{\rm turb}(\ell) \propto \Delta x^{1/3}$ for Kolmogorov turbulence, so turbulent support weakens at finer levels, exacerbating the mismatch.

We therefore conclude that *turbulence alone cannot replace the holdback*; the two mechanisms address different physical regimes and are complementary rather than interchangeable.

9.6 Holdback as a Workaround for Cooling Errors

The analysis above assumes that all Jeans-triggered refinements at high redshift are physically motivated. However, as demonstrated in Section 3.1.1, the standard Newton–Raphson cooling solver can overshoot $T_{\rm eq}$, producing transiently low temperatures that shrink $\lambda_{\rm J}$ by up to an order of magnitude. Similarly, a single-redshift cooling table applied at the wrong epoch can shift $T_{\rm eq}$ by factors of ~2 (Section 3.4). These numerical artefacts generate *spurious* Jeans triggers that dominate the refinement budget at high z.

Table 6. Comparison of one coarse time-step at $z \approx 9$ with and without the holdback mechanism, both using the exact cooling integrator with multi-z table interpolation and $N_J = 4$.

Quantity	Holdback ON	Holdback OFF
Active levels	12–14	12–17
Wall time (s)	125	418
Grid memory fraction	18.3%	24.9%
Stars formed	245	277
Sinks formed	25	25
Fine time-step Δt	3.9×10^{-3}	4.0×10^{-4}
Star formation levels	12–14	12–17

With a correct cooling implementation — the Townsend exact integrator eliminating N-R overshooting, and multi-redshift table interpolation providing the appropriate UV background at each epoch — most diffuse gas at high redshift remains in photoionisation equilibrium at $T \approx 10^4$ K. The Jeans length in this gas is large ($\lambda_J \sim 80$ kpc at $n_H = 0.01\,\mathrm{cm}^{-3}$), well above the refinement threshold, and the Jeans criterion does not trigger. Only genuinely self-shielded, collapsing gas triggers refinement — which is the physically correct behaviour.

Numerical test. We test this hypothesis directly by running two cosmological simulations from $z = 50$ to $z \approx 9$, restarting from identical initial conditions with the Jeans criterion active ($N_J = 4$ at all levels):

(i) **Holdback ON + exact cooling**: standard configuration with the Townsend integrator and multi-z tables;

(ii) **Holdback OFF + exact cooling**: identical, but with `q_refine_holdback = .false.`

Both runs use a 512^3 base grid ($\ell_{\min} = 9$, $\ell_{\max} = 17$) with 12 MPI ranks. Table 6 summarises the results after one coarse time-step at $z \approx 9$.

The key result is that the simulation *without holdback did not undergo a refinement cascade*. Grid memory usage increased by only 36 per cent (24.9% vs. 18.3%), and the total number of sinks (the most massive collapsed objects) remained identical at 25. The 3.3× increase in wall time is entirely attributable to the CFL constraint at level 17 (the fine time-step is ∼10× smaller), not to an explosion in the number of cells. Stars form at level 17 in the holdback-off run — physically appropriate resolution for the densest, self-shielded gas.

These results support the interpretation that the cosmological holdback mechanism has historically been compensating for cooling solver artefacts rather than preventing physically motivated refinement. With an accurate cooling implementation, the holdback is no longer necessary for *stability*; its remaining role is purely one of *computational cost control*.

10 CONCLUSIONS

We have presented three improvements to the handling of the Jeans criterion in cosmological AMR simulations, implemented in the CU-RAMSES code:

(i) **Improved cooling module.** The Eunha cooling module replaces the standard RAMSES Newton–Raphson solver with the Townsend (2009) exact integration scheme applied to 3D Grackle equilibrium cooling tables (Smith et al. 2017). This eliminates temperature overshooting near thermal equilibrium, removes stochastic noise from the Jeans length evaluation, and provides density-dependent metallicity and non-LTE molecular weights. Multi-redshift table interpolation tracks the evolving UV background (Haardt & Madau 2012) across cosmic time, eliminating the systematic bias of fixed-z tables.

(ii) **Jeans-independent holdback bypass.** The `jeans_bypass_holdback` option allows the Jeans criterion to operate ahead of the cosmological holdback, preventing artificial fragmentation and smoothing the star formation rate transition when levels are released. A configurable level limit (`n_jeans_bypass_levels`) and a physics-based density threshold (`d_jeans_thre`) provide complementary safeguards against runaway refinement cascades.

(iii) **Quantitative analysis.** We have shown that differences between the RAMSES and Grackle cooling rates propagate into 40–80 per cent errors in the Jeans length at low densities, and that the N-R temperature overshoot introduces transient refinement noise that the exact integrator eliminates entirely.

(iv) **Holdback diagnosis.** A direct numerical test at $z \approx 9$ demonstrates that with accurate cooling, the cosmological holdback can be disabled without triggering a refinement cascade (Section 9.6). The holdback has historically compensated for cooling artefacts; with the exact integrator and multi-z tables, its role reduces to computational cost control. Turbulent pressure support, while physically important, cannot replace the holdback in the diffuse regime where it is most needed (Section 9.5).

Future work will extend the Jeans criterion to include turbulent pressure support (equation 25), using subgrid-scale turbulence models to estimate the unresolved velocity dispersion on the grid scale. A full star formation rate comparison from $z = 50$ to $z = 1$ between the standard cooling with holdback and the exact cooling without holdback is in progress. This turbulence-dependent refinement criterion, combined with the improved cooling module and holdback bypass presented here, will provide a comprehensive framework for accurate Jeans length resolution in next-generation cosmological simulations.

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